Debadrata Sarkar

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Research Interests

My research interests lie in human-centered robotics, particularly wearable assistive robots and haptics for collaborative robotics. Building on my experience in end-to-end robotic system development, user intent estimation, and human-in-the-loop personalization, I aim to explore embodied intelligence at the intersection of robotics and machine learning, with an emphasis on real-world deployment. Specifically, I seek to develop intelligent control policies for wearable robots that integrate environmental perception with proprioceptive sensing, adapt across tasks and environments, and continuously co-adapt with human users.

Research output: 4 journal articles, with 3 more under review • 15 conference papers • 3 patent applications (1 U.S.)

Education

National Institute of Technology Durgapur 2015 – 2019

B.Tech. in Mechanical Engineering *CGPA: 9.30/10 | Ranked 5th of 126 (top 4%)*

B.Tech. project with Prof. Shibendu Shekhar Roy: *A Soft Bending-type Actuator using Hyper-elastic Materials: Development, Analysis and Characterization* – **Best Paper Award**, Conf. on Advances in Robotics (AIR 2019).

Research Experience

CSIR-Central Mechanical Engineering Research Institute (CMERI) May 2026 – Present

Research Project Scientist | *Human-Centred Robotics and Cybernetics Group* *Durgapur, India* • *funded by ICMR*

- **XoNav: hip exoskeleton as a personalized haptic interface for assisted navigation** [U1] [Video]
  - Sagittal- and frontal-plane quasi-direct-drive actuators of a four-DoF exoskeleton, under joint impedance control, render kinesthetic navigation cues at the hip, synchronized to the gait cycle by LSTM-based *gait-state estimation* from two thigh IMUs that generalizes to unseen users.
  - Each wearer's cues are tuned by *constrained, preference-based human-in-the-loop (HIL) Bayesian optimization* combining a detectability constraint, analytic safety screening and pairwise comfort comparisons. Blindfolded Wizard-of-Oz trials achieved 98.8% corridor adherence, and *visuo-inertial SLAM* (RTAB-Map) with the Nav2 navigation stack enabled autonomous guidance.
- **Tread: visuo-inertial terrain- and locomotion-context-aware gait estimation** [U2] [Code] [Video]
  - A causal perception chain fuses a chest-mounted camera with four lower-limb IMUs and runs onboard edge inference over ROS 2 and micro-ROS. Upcoming terrain, recognized in advance by *transfer learning* on a frozen ResNet backbone, gates LSTM experts that identify the locomotion mode with fidget rejection.
  - Instead of hard switching, the mode posterior enters a *shared LSTM gait encoder* as a *probabilistic context*, and multi-task heads estimate continuous bilateral gait phase and walking speed; orientation-based features keep the models robust to sensor mounting, and the terrain-gated hierarchy combines datasets from separate protocols.
- **ExoStack: open-source modular ROS 2 architecture for wearable robots** [Code] [Docs] (under review)
  - An ESP32/micro-ROS *hardware abstraction* layer adds sensors and actuators from templates (declarative descriptors, self-registering drivers), and *plug-in ROS 2 nodes* add synchronization, gait-event detection, assistance profiles, motion planning and cost evaluation, so it scales to multiple joints and reconfigures for new wearable robots without core-code changes. One profile interface serves deployment and HIL optimization (BO, CMA-ES) on knee and hip exosuits.

Indian Institute of Technology Delhi Sep 2024 – Jan 2026

Senior Project Scientist | *Department of Applied Mechanics* *New Delhi, India* • *funded by DRDO*

- **Upper-limb muscle power augmentation with BMI**
  - Led full-stack development of a tendon-driven elbow-flexion exosuit, from mechanical design, tendon transmission, sensing and fabrication to embedded control on an NVIDIA Jetson, and implemented *biosignal-based admittance control* for gravity compensation, tuned in load-lifting trials.
- **Lower-limb assistance and gait-laboratory testbed**
  - Developed a tendon-driven ankle exoskeleton that executes plantarflexion profiles derived from offline *biomechanical simulation*, triggered by FSR-based gait-event detection, and a *HIL testbed* that synchronizes exosuit sensing with an instrumented treadmill, motion capture, EMG and a metabolic analyzer.

CSIR-Central Mechanical Engineering Research Institute (CMERI) Jul 2019 – Sep 2023

Project Assistant | *Robotics and Automation Group* *Durgapur, India* • *funded by DRDO*

Core member of a soft-exosuit project from inception: actuators, sensing, embedded systems, control, test rigs, human trials

- **SoftEXOwalk: human-in-the-loop optimized soft exosuit for knee-flexion assistance** [J1]
  - An untethered bilateral soft exosuit with textile force transmission runs on ROS with GPU edge inference; its *self-sensing pSTAMs* feed back transmitted force, estimated from embedded strain sensing by a data-driven LSTM without external force sensors (patent pending), and walking-speed estimation enables speed-adaptive control.

- Splitting transmitted force by gait phase defines a *custom metabolic-free objective* for rapid, *constrained HIL Bayesian optimization* that personalizes assistance in about 3 min per speed, evaluated with multi-subject sEMG.

- **Soft haptic glove for VR and telerobotics** [J2]
  - Custom flexible bend sensors track the fingers, and *multimodal haptic rendering* pairs cutaneous cues from pneumatic fingertip tactile arrays (contact location, surface features) with motor–tendon kinesthetic feedback of variable stiffness (size, shape, stiffness).
  - The glove drives Unity-based VR and virtual teleoperation with collider-based contact rendering over ROS, was evaluated *psychophysically* (discrimination tasks, NASA-TLX) and later teleoperated an Allegro hand.
- **Soft actuation and sensing**
  - Developed pSTAM low-pressure PAMs [P2] from design, fabrication and static and dynamic characterization to data-driven neural-surrogate modeling and design optimization (open-source *pSTAM design toolbox* [Code]) and *NN–PID hybrid control*; sensorized pSTAMs embed conductive-elastomer, liquid-metal and conductive-fabric sensing.
  - Designed a *two-stage pneumatic supply* [P1] via a data-driven model for untethered wearables and aerial manipulators.
- **Indo–Korea Joint Network Centre on Robotics**
  - Developed an O-ring-reinforced soft steering head for endoscopy, designed through FEA [C2].
  - Built an aerial manipulator that combines a quadrotor, a 2-DoF arm and a pneumatic soft gripper [C1].

The BioRobotics Institute, Scuola Superiore Sant'Anna May 2019 – Jul 2019
Visiting Researcher | *Neuro-Robotic Touch Laboratory, with Prof. Calogero M. Oddo* *Pisa, Italy*

- **Pneumatic fingertip tactile display for surgical haptic feedback** [C3]
  - Designed a 3×3 elastomeric tactor array, guided by FEA, to give haptic feedback to laparoscopic surgeons, and characterized spatial stimulus perception through psychophysical experiments.

Awards and Honors

Best Paper Award, AIR 2019, IIT Madras • **Best Poster Award**, AIR 2023, IIT Ropar

Technical Skills

Mechanical design: SolidWorks; hyperelastic FEA (Abaqus, COMSOL); tendon and textile transmissions; silicone casting
Electronics & embedded: NVIDIA Jetson, ESP32, Teensy; KiCad PCB design; CAN-bus actuators; electro-pneumatics
Software: ROS 2, micro-ROS, Nav2, RTAB-Map, Unity, MuJoCo; Python, C/C++, MATLAB; Kinova Gen3, Allegro hand
Learning & optimization: LSTM, CNN, transfer learning, neural surrogates; Bayesian optimization, CMA-ES, PSO
Human trials: Bertec treadmill, COSMED, Delsys EMG, Xsens, OptiTrack, PhaseSpace, HTC Vive; psychophysics

Selected Publications *(complete list on Google Scholar)*

[J1] **D. Sarkar**, S. Chakraborty, S. Sen, S. S. Roy, and A. Arora, "SoftEXOwalk: A Human-in-the-Loop optimized Soft Exosuit for Assisting Knee Flexion using Sensorized Pneumatic Muscles," *IEEE Trans. Med. Robot. Bionics*, 2026. doi:10.1109/TMRB.2026.3722336

[J2] **D. Sarkar**, A. Kumar, and A. Arora, "A Soft Glove with Proprioceptive Sensing and Multi-modal Haptic Feedback for VR and Telerobotic Applications," *J. Bionic Eng.*, vol. 22, pp. 684–702, 2025. doi:10.1007/s42235-024-00642-z

[U1] **D. Sarkar**, S. Sen, and A. Arora, "XoNav: An Exoskeleton as a Personalized Haptic Interface for Assisted Navigation," under review, *IEEE Robotics and Automation Letters*, 2026. arXiv:2610.27271

[U2] **D. Sarkar**, A. Bachhar, and A. Arora, "Tread: A Visuo-Inertial Framework for Terrain and Locomotion Context-Aware Gait-Phase and Speed Estimation in Wearable Robotics," under review, *IEEE Sensors J.*, 2026. arXiv:2610.19067

Patent Applications

[P1] A. Arora, **D. Sarkar**, and S. Sen, "Two-stage pneumatic supply architecture for light-weight untethered pneumatic actuation systems," U.S. Patent Appl. 18/604,087 (pub. US 2024/0309891 A1); Indian Appl. 202311017238; pending.

[P2] A. Arora, **D. Sarkar**, S. Sen, and C. Har, "Custom made silicone bladder based low pressure actuated pneumatic artificial muscle," Indian Patent Appl. 202211021971 (2022); pending.

Selected Conference Publications

[C1] **D. Sarkar**, A. Arora, S. Sen, S. S. S. Katta, D. Shashank, M. Rohan, and S. K. Saha, "Development of an Autonomous UAV Integrated with a Manipulator and a Soft Gripper," *13th Asian Control Conference (ASCC)*, **Jeju, South Korea**, 2022. doi:10.23919/ASCC56756.2022.9828332

[C2] **D. Sarkar**, S. Chakraborty, A. Arora, and S. Sen, "Ring Reinforced Silicone based Steering Head for Endoscopy like Applications: FEM simulation, development and force characterization," *21st International Conference on Control, Automation and Systems (ICCAS)*, **Jeju, South Korea**, 2021. doi:10.23919/ICCAS52745.2021.9649865

[C3] **D. Sarkar**, J. D'Abbraccio, D. Camboni, L. Massari, A. Arora, and C. M. Oddo, "A Pneumatic Haptic Display for Collaborative Robotics applications," *IEEE International Workshop on Metrology for Industry 4.0 & IoT*, **Rome, Italy**, 2020. doi:10.1109/MetroInd4.0IoT48571.2020.9138283